CLEAN Algorithm for Imaging Quality Enhancement

S. N. Sur, S. Bera, A. Anand and S. Rai

Abstract In ITS, radar plays an important role. Detection of the target and characterization is very important. Similarly, in dynamic channel condition, target image quality degrades considerable due to sidelobe enhancement and defocusing. The clean algorithm is introduced to eliminate sidelobes and significantly improves the target detection performance of radar signals. The experimental results demonstrate the effectiveness of the CLEAN algorithm.

Keywords Clean algorithm · RADAR · Inverse synthetic aperture radar (ISAR) · Range · Correlation · Spectrum

1 Introduction

Automobile radar is one of highly growing industries due to increase in the heavy traffic worldwide. Along with the increase with the on-road traffic, safety has become very important. Also, traffic congestion is another aspect to look into. Hence, the introduction of intelligent transportation system (ITS) in highway network is of utmost significance [1]. One of the solutions to the above problems is remote sensing [2, 3] the different obstacles and act accordingly. Remote sensing is again a challenging job under ITS due to highly dynamic channel condition. In dynamic channel situation, the nearby or adjacent targets suffer from increase in sidelobe and thus overlapping on each other causing target peak misallocation [4, 5]. So, to meet the demand, very high-resolution imaging radar needs to be developed and this paper deals with the development of such high-resolution imaging radar for vehicular application so as to recognize and classify targets or obstacles clearly and improving the on-road traffic conditions [6–8].

The requirement of high-resolution radar image so as to identify the targets clearly a technique known as CLEAN [4, 9, 10] may be employed which has the capability

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of reducing the sidelobes enhancement thus lowering the chances to target overlapping. Hence, the application of this algorithm will lead to proper localization and classification of the closely spaced vehicular targets so as to meet the challenge of heavy traffic. In this paper, authors have introduced a hybrid processing scheme for a polyphase coded spread spectrum system by combining the matched filter and coherent clean algorithm in order to enhance the performance of the radar.

2 Clean Algorithm

In this paper, the CLEAN algorithms are used to improve the image quality of polyphase coded spread spectrum ISAR radar. The practical result demonstrates the effectiveness of the CLEAN algorithm. Here, signal processing methods, like cross correlation, FFTs, and CLEAN algorithm on target images, are incorporated to improve the radar system performance.

The algorithm, as in Fig. 1, initially brightest spot in the image map (D) has been identified. The location and intensity information for the brightest spot are retained in a clean map (G). Once the brightest spot is removed from the image map (D), the procedure is repeated with the next brightest source in the image map until some predetermined condition has been reached. For the spot identification, peak detection or blind spot detection algorithm is used to prepare a hot spot imaging.

3 Experimental Results

Here, for the sensing purpose, polyphase coded signal has been used and has 100 MHz bandwidth. Arbitrary wave generator (AWG) is used to generate IF signal over 70 MHz. For increasing the radar range resolution RF frequency swept over the frequency rage 1–2 GHz. For the target characterization toy car has been placed over a 2-axis positioner. The target has the dimension of 0.6 m (cross range) $\times$ 0.9 (range). For range cross range 2D imaging we have rotated the target in the azimuth direction. The hart of the receiver is vector signal analyzer (VSA), which is a programmable device. All the receiver signal processing algorithms have been implemented in the VSA. The down-converted signal is used to extract the target information using VSA.

The radar signal processing (RSP) algorithm includes a correlation process for ranging information followed by 2D FFT for cross range information. The CLEAN algorithm has been applied to improve the image quality.

Figure 2 represents image of the target without background cancellation. As in the figure, the target is detected but with back scattered false target. This additional hot spot results in false detection and needs to be removed.

Whereas, Fig. 3 represents image of the target after background cancellation. As in figure, background cancellation effectively able to remove the back scatters.

Effectiveness of the CLEAN algorithm is depicted in Fig. 4. By comparing

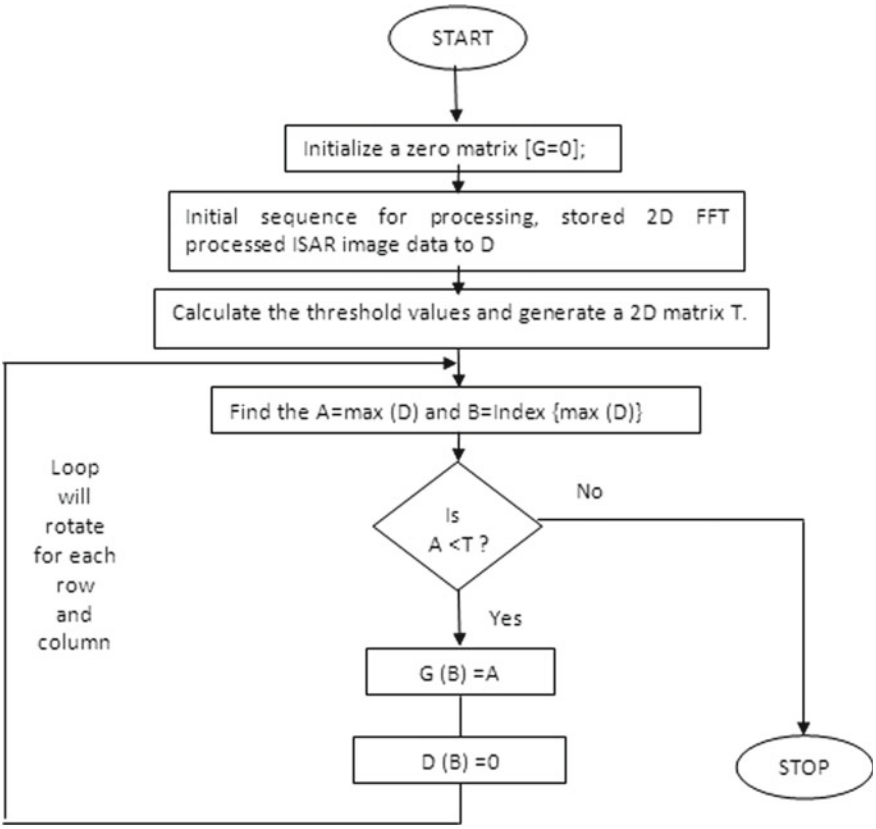

Fig. 1 CLEAN algorithm for image cleaning

Figs. 3 and 4 one can easily figure out the impact of the CLEAN algorithm. CLEAN algorithm helps to create more distinct image and thereby improves the image quality.

4 Conclusion

This paper demonstrates the CLEAN algorithm and its effectiveness for creating hot spot image of the target. In critically danse conditions or during the imaging of the contiguous target, the image quality degrades due to sidelobe enhancement. And in this regard CLEAN algorithm is a very good tool to improve the image quality. As depicted in the result above it can be seen that the efficient use of CLEAN algorithm provides significant enhancement in the performance of the system.

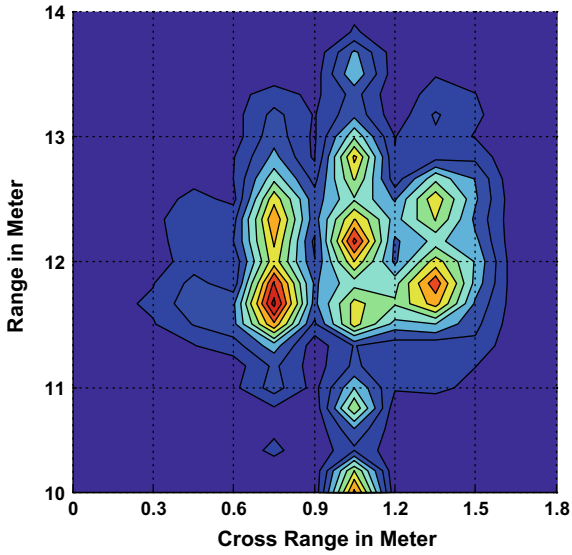

Fig. 2 Target (car) image before background cancellation

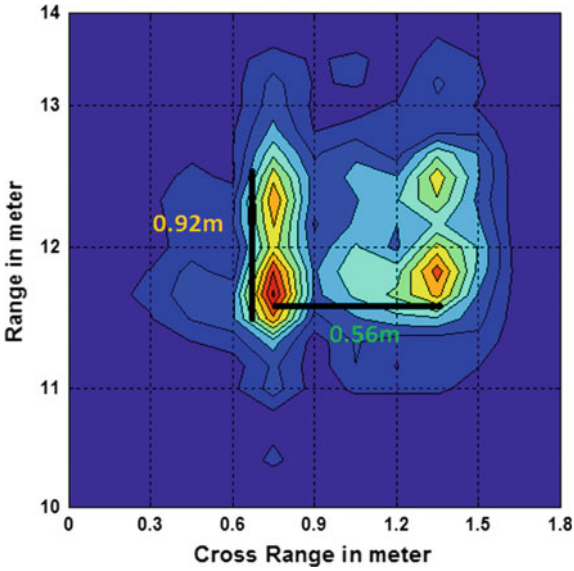

Fig. 3 Target (car) image after background cancellation

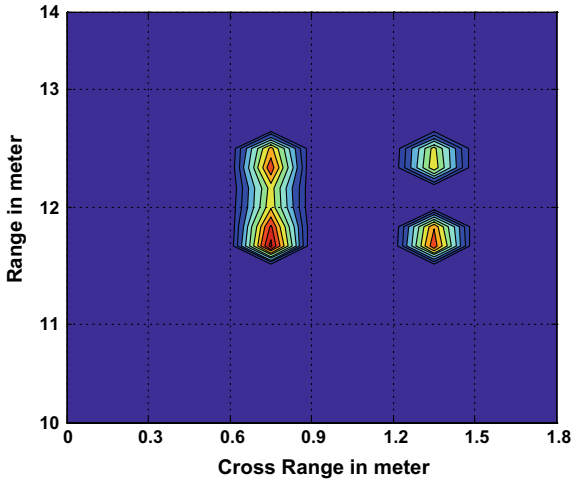

Fig. 4 Cleaned target response

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